

How Algebra Sees Combinatorial Structure.

대수는 어떻게 조합적 구조를 바라보는가

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MIMIC Seminar

What is Combinatorics?

Counting # Combinatorial objects.

What is Combinatorics?

Counting # Combinatorial objects.

↓ examples.

Young Tableaux

1	2	3
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1	2
3	

1	3
2	

1
2
3

Integer Partitions

4

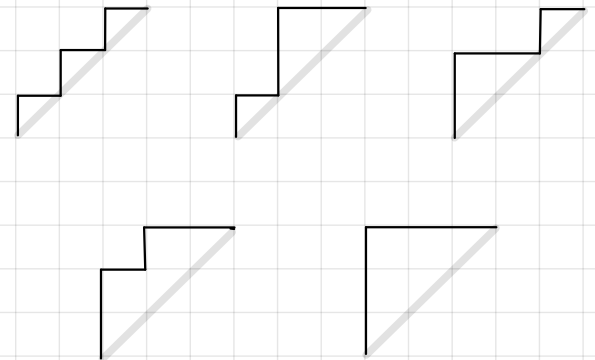
$3+1$

$2+2$

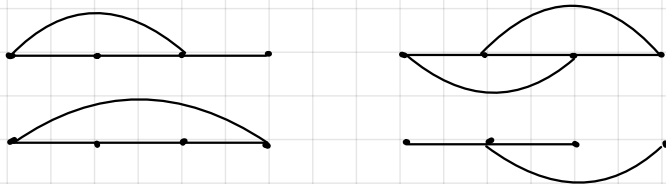
$2+1+1$

$1+1+1+1$

Dyck Paths



Graphs



Permutations

1 2 3 4
1 2 4 3
1 3 2 4

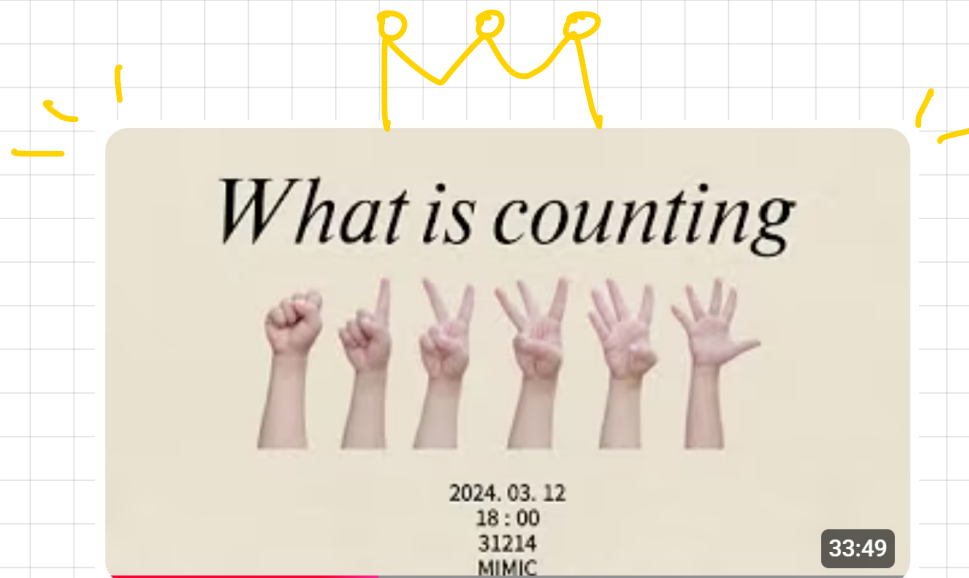
⋮

How...?

- $\left(\begin{array}{c} \text{Combinatorial} \\ \text{Object 1} \end{array} \right) \xleftrightarrow{\text{bijection}} \left(\begin{array}{c} \text{Combinatorial} \\ \text{Object 2} \end{array} \right)$
- $\left(\begin{array}{c} \# \text{Objects} \\ \text{that we want to know} \end{array} \right) \xleftrightarrow{\text{bijection}} \left(\begin{array}{c} \# \text{Objects} \\ \text{that we already know} \end{array} \right)$

How...?

- $\left(\begin{array}{c} \text{Combinatorial} \\ \text{Object 1} \end{array} \right) \xleftrightarrow{\text{bijection}} \left(\begin{array}{c} \text{Combinatorial} \\ \text{Object 2} \end{array} \right)$
- $\left(\begin{array}{c} \# \text{Objects} \\ \text{that we want to know} \end{array} \right) \xleftrightarrow{\text{bijection}} \left(\begin{array}{c} \# \text{Objects} \\ \text{that we already know} \end{array} \right)$



← Presented by
Hojoon Lee

셈이란 무엇인가
조회수 159회 • 2년 전

For example ...

$$4 = 3 + 1$$

$$= 2 + 2$$

$$= 2 + 1 + 1$$

$$= 1 + 1 + 1 + 1$$

$$5 = 4 + 1$$

$$= 3 + 2$$

$$= 3 + 1 + 1$$

$$= 2 + 2 + 1$$

$$= 2 + 1 + 1 + 1$$

$$= 1 + 1 + 1 + 1 + 1$$

For example ...

$$\begin{aligned} 4 &= 3 + 1 \\ &= 2 + 2 \quad \leftarrow \text{part} \\ &= 2 + 1 + 1 \\ &= 1 + 1 + 1 + 1 \end{aligned}$$

$$\begin{aligned} 5 &= 4 + 1 \\ &= 3 + 2 \\ &= 3 + 1 + 1 \\ &= 2 + 2 + 1 \\ &= 2 + 1 + 1 + 1 \\ &= 1 + 1 + 1 + 1 + 1 \end{aligned}$$

→ We call it "integer partition".

For example ...

$$4 = 3 + 1$$

$$= 2 + 2$$

$$= 2 + 1 + 1$$

$$= 1 + 1 + 1 + 1$$

- Distinct parts.

$$4, 3 + 1$$

- Odd parts.

$$3 + 1, 1 + 1 + 1 + 1$$

For example ...

$$5 = 4 + 1$$

$$= 3 + 2$$

$$= 3 + 1 + 1$$

$$= 2 + 2 + 1$$

$$= 2 + 1 + 1 + 1$$

$$= 1 + 1 + 1 + 1 + 1$$

- Distinct parts.

$$5, 4 + 1, 3 + 2$$

- Odd parts.

$$5, 3 + 1 + 1,$$

$$1 + 1 + 1 + 1 + 1$$

For example ...

$$5 = 4 + 1$$

$$= 3 + 2$$

$$= 3 + 1 + 1$$

$$= 2 + 2 + 1$$

$$= 2 + 1 + 1 + 1$$

$$= 1 + 1 + 1 + 1 + 1$$

- Distinct parts.

$$5, 4 + 1, 3 + 2$$

- Odd parts.

$$5, 3 + 1 + 1,$$

$$1 + 1 + 1 + 1 + 1$$

→ Observation : (# Distinct parts) = (# Odd parts)

For example ...

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$



Define a bijection

For example ...

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$

Define a bijection

$$\begin{array}{c} 4 \\ 3+1 \end{array}$$

?

$$\begin{array}{c} 3+1 \\ 1+1+1+1 \end{array}$$

$$\begin{array}{c} 5 \\ 4+1 \\ 3+2 \end{array}$$

?

$$\begin{array}{c} 5 \\ 3+1+1 \\ 1+1+1+1+1 \end{array}$$

Generating Function

Integer Partition $\longrightarrow$ Generating Function

$$4$$

$$= 3 + 1$$

$$= 2 + 2$$

$$= 2 + 1 + 1$$

$$= 1 + 1 + 1 + 1$$

$$\lambda^4 = \lambda^4$$

$$\lambda^{3+1} = \lambda^3 \cdot \lambda$$

$$\lambda^{2+2} = \lambda^2 \cdot \lambda^2$$

$$\lambda^{2+1+1} = \lambda^2 \cdot \lambda \cdot \lambda$$

$$\lambda^{1+1+1+1} = \lambda \cdot \lambda \cdot \lambda \cdot \lambda$$

Generating Function

Integer Partition $\longrightarrow$ Generating Function

$$4$$

$$= 3 + 1$$

$$= 2 + 2$$

$$= 2 + 1 + 1$$

$$= 1 + 1 + 1 + 1$$

$$x^4 = x^4$$

$$x^{3+1} = x^3 \cdot x$$

$$x^{2+2} = x^2 \cdot x^2$$

$$x^{2+1+1} = x^2 \cdot x \cdot x$$

$$x^{1+1+1+1} = x \cdot x \cdot x \cdot x$$

$\downarrow$ Sum

$$5x^4 = (\# \text{partition of } 5) \cdot x^5$$

Generating Function

Integer Partition $\longrightarrow$ Generating Function

n

$(\# \text{ partition of } n) \cdot x^n$

$\downarrow$ Sum for all n .

$$\sum_{n \geq 0} (\# \text{ partition of } n) \cdot x^n$$

def) Given a sequence $(a_n)_{n \geq 0}$, the formal power series

$$A(x) = \sum_{n=0}^{\infty} a_n x^n$$

is called the generating function of the sequence.

Generating Function

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$



Define a bijection

Generating Function

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$

$$\text{G.F. of distinct parts} \stackrel{?}{=} \text{G.F. of odd parts.}$$

$$\sum_{n \geq 0} (\# \text{ Distinct parts for } n)$$

$$\sum_{n \geq 0} (\# \text{ Odd parts for } n)$$

Generating Function

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$

$$\text{G.F. of distinct parts} \stackrel{?}{=} \text{G.F. of odd parts.}$$

$$\sum_{n \geq 0} (\# \text{ Distinct parts for } n)$$

$$\sum_{n \geq 0} (\# \text{ Odd parts for } n)$$

Generating Function

$$(1+x)(1+x^2)(1+x^3) = \underbrace{(x^0+x^1)}_{\text{Choose 0 or 1}} \underbrace{(x^0+x^2)}_{\text{Choose 0 or 2}} \underbrace{(x^0+x^3)}_{\text{Choose 0 or 3}}$$

$$\text{e.g. } x \cdot x^2 \cdot 1 = x^{1+2} \longleftrightarrow \text{Partition of 3 ; } 1+2$$

$$1 \cdot 1 \cdot x^3 = x^3 \longleftrightarrow \text{Partition of 3 ; } 3$$

$$x \cdot 1 \cdot x^3 = x^{1+3} \longleftrightarrow \text{Partition of 4 ; } 1+3$$

$\vdots$

$$\therefore \sum_{n \geq 0} (\# \text{Distinct parts for } n) = (1+x)(1+x^2)(1+x^3)(1+x^4) \dots$$

$$= \prod_{n \geq 1} (1+x^n)$$

Generating Function

$$(1+x)(1+x^2)(1+x^3) = \underbrace{(x^0 + x^1)}_{\text{Choose 0 or 1}} \underbrace{(x^0 + x^2)}_{\text{Choose 0 or 2}} \underbrace{(x^0 + x^3)}_{\text{Choose 0 or 3}}$$

$$\text{e.g. } x \cdot x^2 \cdot 1 = x^{1+2} \longleftrightarrow \text{Partition of 3 ; } 1+2$$

$$1 \cdot 1 \cdot x^3 = x^3 \longleftrightarrow \text{Partition of 3 ; } 3$$

$$x \cdot 1 \cdot x^3 = x^{1+3} \longleftrightarrow \text{Partition of 4 ; } 1+3$$

$\vdots$

$$\begin{aligned} \therefore \sum_{n \geq 0} (\# \text{Distinct parts for } n) &= (1+x)(1+x^2)(1+x^3)(1+x^4) \dots \\ &= \prod_{n \geq 1} (1+x^n) \end{aligned}$$

Generating Function

$$(1+x)(1+x^2)(1+x^3) = \underbrace{(x^0+x^1)}_{\text{Choose 0 or 1}} \underbrace{(x^0+x^2)}_{\text{Choose 0 or 2}} \underbrace{(x^0+x^3)}_{\text{Choose 0 or 3}}$$

e.g. $x \cdot x^2 \cdot 1 = x^{1+2} \longleftrightarrow \text{Partition of 3 ; } 1+2$

$1 \cdot 1 \cdot x^3 = x^3 \longleftrightarrow \text{Partition of 3 ; } 3$

$x \cdot 1 \cdot x^3 = x^{1+3} \longleftrightarrow \text{Partition of 4 ; } 1+3$

$\vdots$

Choose 0 or n .

$$\begin{aligned} \therefore \sum_{n \geq 0} (\# \text{Distinct parts for } n) &= (1+x)(1+x^2)(1+x^3)(1+x^4) \cdots \\ &= \prod_{n \geq 1} (1+x^n) \end{aligned}$$

Generating Function

$$\text{For odd } 1 : (x^{0 \cdot 1} + x^{1 \cdot 1} + x^{2 \cdot 1} + x^{3 \cdot 1} + \dots) = (1 + x + x^2 + x^3 + \dots) \\ = \frac{1}{1-x}$$

$$\text{For odd } 3 : (x^{0 \cdot 3} + x^{1 \cdot 3} + x^{2 \cdot 3} + x^{3 \cdot 3} + \dots) = (1 + x^3 + x^6 + x^9 + \dots) \\ = \frac{1}{1-x^3}$$

⋮

$$\text{For odd } k : (x^{0 \cdot k} + x^{1 \cdot k} + x^{2 \cdot k} + \dots) = \frac{1}{1-x^k}$$

$$\therefore \sum_{n \geq 0} (\# \text{ Odd parts for } n) = \prod_{k: \text{ odd}} \frac{1}{1-x^k} = \prod_{m \geq 1} \frac{1}{1-x^{2m-1}}$$

Generating Function

$$\text{For odd } 1 : (x^{0 \cdot 1} + x^{1 \cdot 1} + x^{2 \cdot 1} + x^{3 \cdot 1} + \dots) = (1 + x + x^2 + x^3 + \dots) \\ = \frac{1}{1-x}$$

$$\text{For odd } 3 : (x^{0 \cdot 3} + x^{1 \cdot 3} + x^{2 \cdot 3} + x^{3 \cdot 3} + \dots) = (1 + x^3 + x^6 + x^9 + \dots) \\ = \frac{1}{1-x^3}$$

⋮

$$\text{For odd } k : (x^{0 \cdot k} + x^{1 \cdot k} + x^{2 \cdot k} + \dots) = \frac{1}{1-x^k}$$

$$\therefore \sum_{n \geq 0} (\# \text{ Odd parts for } n) = \prod_{k: \text{ odd}} \frac{1}{1-x^k} = \prod_{m \geq 1} \frac{1}{1-x^{2m-1}}$$

Generating Function

$$\text{For odd 1 : } (x^{0 \cdot 1} + x^{1 \cdot 1} + x^{2 \cdot 1} + x^{3 \cdot 1} + \dots) = (1 + x + x^2 + x^3 + \dots) \\ = \frac{1}{1-x}$$

$$\text{For odd 3 : } (x^{0 \cdot 3} + x^{1 \cdot 3} + x^{2 \cdot 3} + x^{3 \cdot 3} + \dots) = (1 + x^3 + x^6 + x^9 + \dots) \\ = \frac{1}{1-x^3}$$

⋮

how many times
will you use k

$$\text{For odd } k : (x^{0 \cdot k} + x^{1 \cdot k} + x^{2 \cdot k} + \dots) = \frac{1}{1-x^k}$$



$$\therefore \sum_{n \geq 0} (\# \text{ Odd parts for } n) = \prod_{k: \text{ odd}} \frac{1}{1-x^k} = \prod_{m \geq 1} \frac{1}{1-x^{2m-1}}$$

Generating Function

$$\text{For odd } 1 : (x^{0 \cdot 1} + x^{1 \cdot 1} + x^{2 \cdot 1} + x^{3 \cdot 1} + \dots) = (1 + x + x^2 + x^3 + \dots) \\ = \frac{1}{1-x}$$

$$\text{For odd } 3 : (x^{0 \cdot 3} + x^{1 \cdot 3} + x^{2 \cdot 3} + x^{3 \cdot 3} + \dots) = (1 + x^3 + x^6 + x^9 + \dots) \\ = \frac{1}{1-x^3}$$

⋮

$$\text{For odd } k : (x^{0 \cdot k} + x^{1 \cdot k} + x^{2 \cdot k} + \dots) = \frac{1}{1-x^k}$$

$$\therefore \sum_{n \geq 0} (\# \text{ Odd parts for } n) = \prod_{k: \text{odd}} \frac{1}{1-x^k} = \prod_{m \geq 1} \frac{1}{1-x^{2m-1}}$$

Generating Function

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$

$$\text{G.F. of distinct parts} \stackrel{?}{=} \text{G.F. of odd parts.}$$

$$\sum_{n \geq 0} (\# \text{ Distinct parts for } n) x^n$$

$$\sum_{n \geq 0} (\# \text{ Odd parts for } n) x^n$$

Generating Function

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$

$$\text{G.F. of distinct parts} \stackrel{?}{=} \text{G.F. of odd parts.}$$

$$\sum_{n \geq 0} (\# \text{ Distinct parts for } n) x^n$$

$$\prod_{k \geq 1} (1 + x^k)$$

Use k or not

$$\sum_{n \geq 0} (\# \text{ Odd parts for } n) x^n$$

$$\prod_{m \geq 1} \frac{1}{1 - x^{2m-1}}$$

Determine how many times
will we use odd m .

Generating Function

$$(\# \text{ Distinct parts}) \stackrel{?}{=} (\# \text{ Odd parts})$$

$$\text{G.F. of distinct parts} \stackrel{?}{=} \text{G.F. of odd parts.}$$

$$\sum_{n \geq 0} (\# \text{ Distinct parts for } n) x^n$$

$$\sum_{n \geq 0} (\# \text{ Odd parts for } n) x^n$$

$$\prod_{k \geq 1} (1 + x^k)$$

$$\stackrel{?}{=}$$

$$\prod_{m \geq 1} \frac{1}{1 - x^{2m-1}}$$

Use k or not

Determine how many times
will we use odd m .

Generating Function

$$\sum_{n \geq 0} (\# \text{Distinct parts for } n) x^n = \prod_{k \geq 1} (1 + x^k)$$

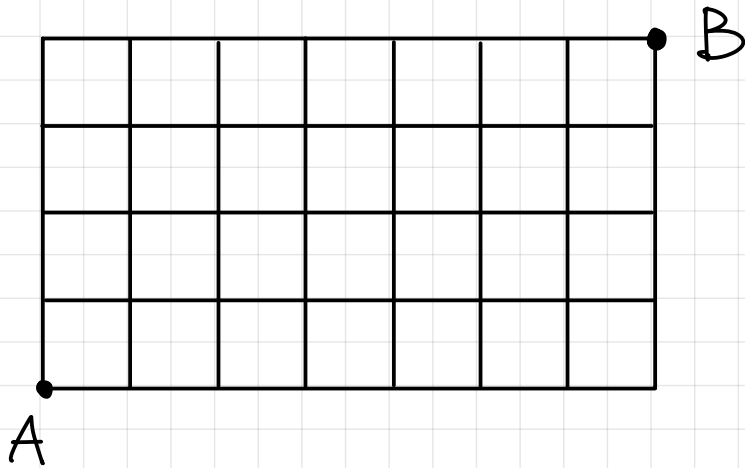
Generating Function

$$\begin{aligned}\sum_{n \geq 0} (\# \text{Distinct parts for } n) x^n &= \prod_{k \geq 1} (1 + x^k) \\&= \prod_{k \geq 1} \frac{(1 + x^k)(1 - x^k)}{1 - x^k} \\&= \prod_{k \geq 1} \frac{1 - x^{2k}}{1 - x^k} \\&= \frac{(1 - x^2)(1 - x^4) \dots}{(1 - x)(1 - x^2)(1 - x^3)(1 - x^4) \dots}\end{aligned}$$

Generating Function

$$\begin{aligned}\sum_{n \geq 0} (\# \text{Distinct parts for } n) x^n &= \prod_{k \geq 1} (1 + x^k) \\&= \prod_{k \geq 1} \frac{(1 + x^k)(1 - x^k)}{1 - x^k} \\&= \prod_{k \geq 1} \frac{1 - x^{2k}}{1 - x^k} \\&= \frac{\cancel{(1 - x^2)} \cancel{(1 - x^4)} \dots}{(1 - x) \cancel{(1 - x^2)} (1 - x^3) \cancel{(1 - x^4)} \dots} \\&= \prod_{m \geq 1} \frac{1}{1 - x^{2m-1}} \\&= \sum_{n \geq 0} (\# \text{Odd parts for } n) x^n\end{aligned}$$

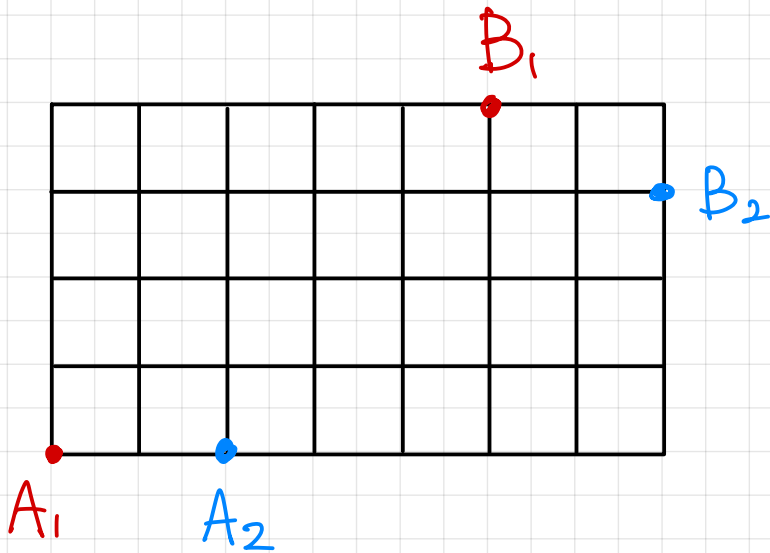
Lattice Path



$$\binom{11}{4} = 11C4 \text{ paths}$$

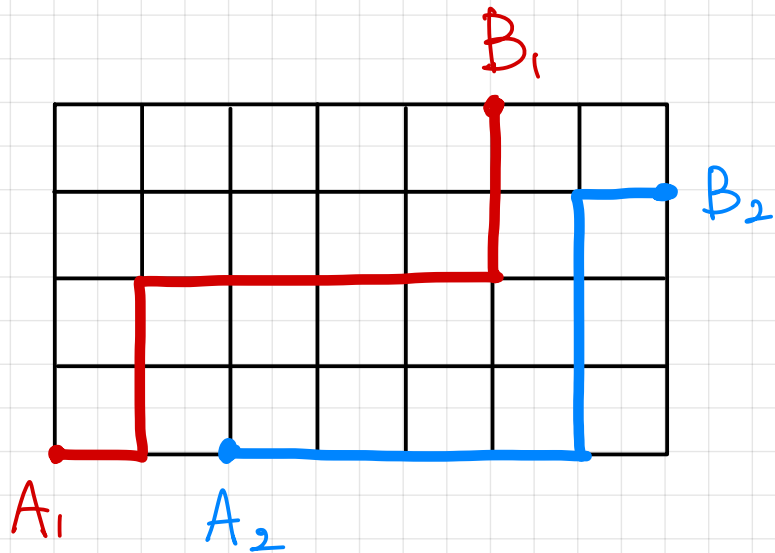
Lattice Path

Draw the two paths so that they do not intersect each other.

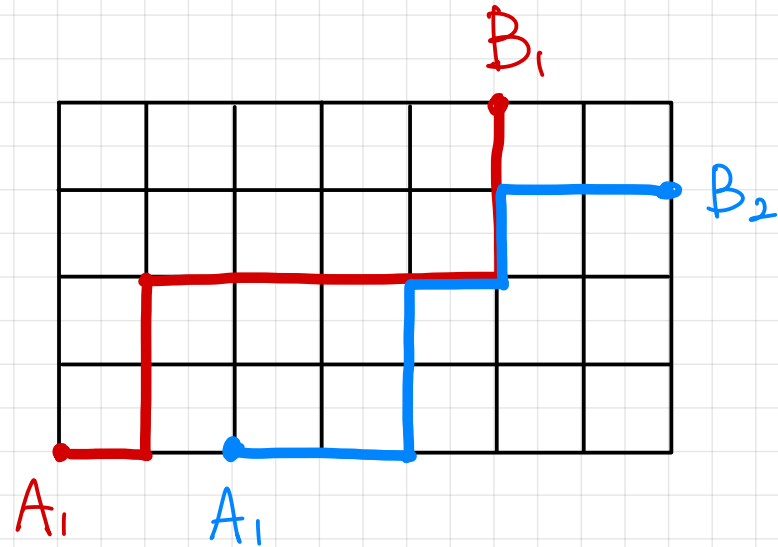


Lattice Path

Draw the two paths so that they do not intersect each other.



→ OK.



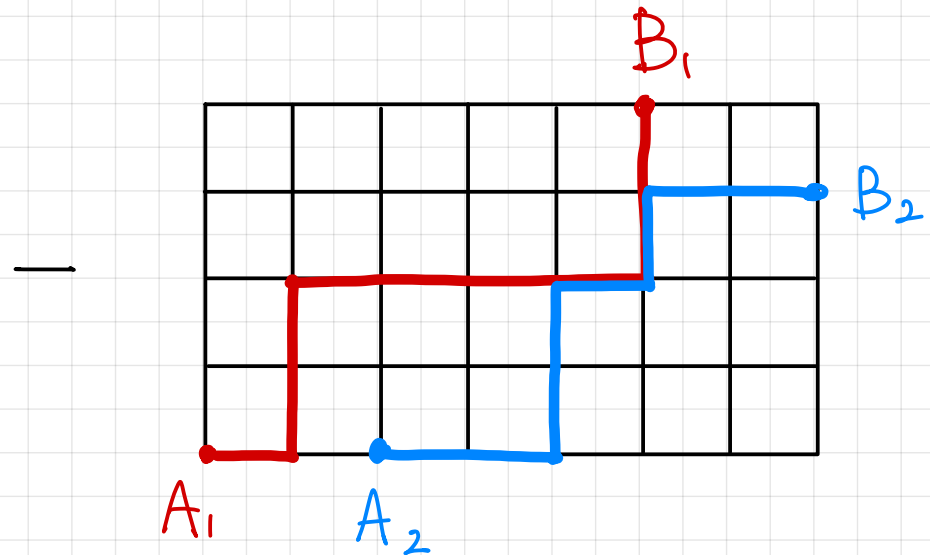
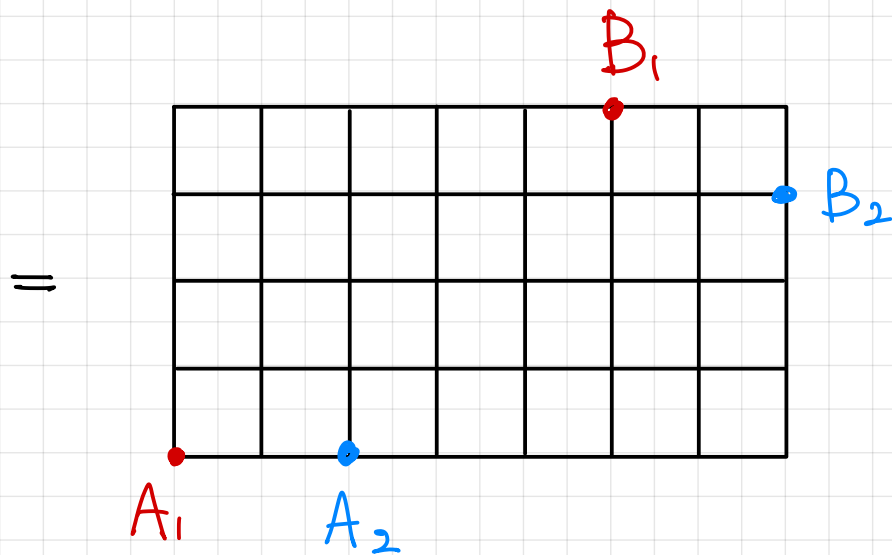
→ Not OK...

Q. How many such pairs?

Lattice Path

Q. How many such pairs?

A. (# Total pairs) - (# Intersecting pairs)



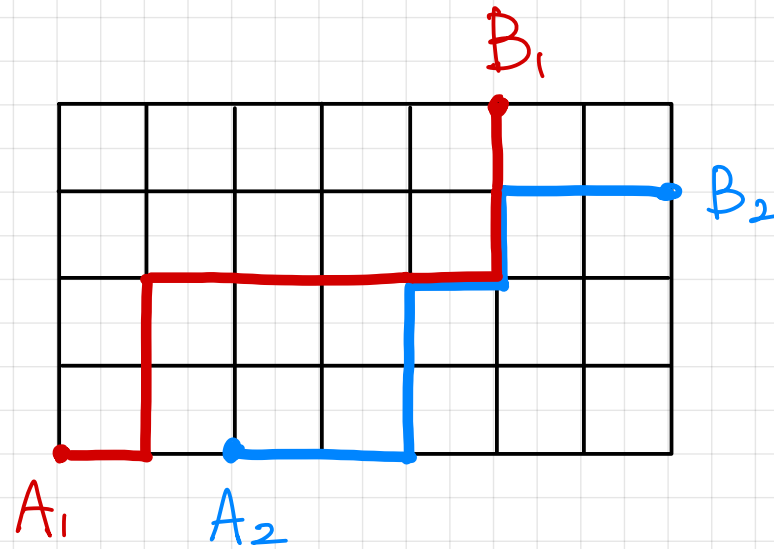
=

$$\binom{4+5}{4} \times \binom{3+5}{3}$$

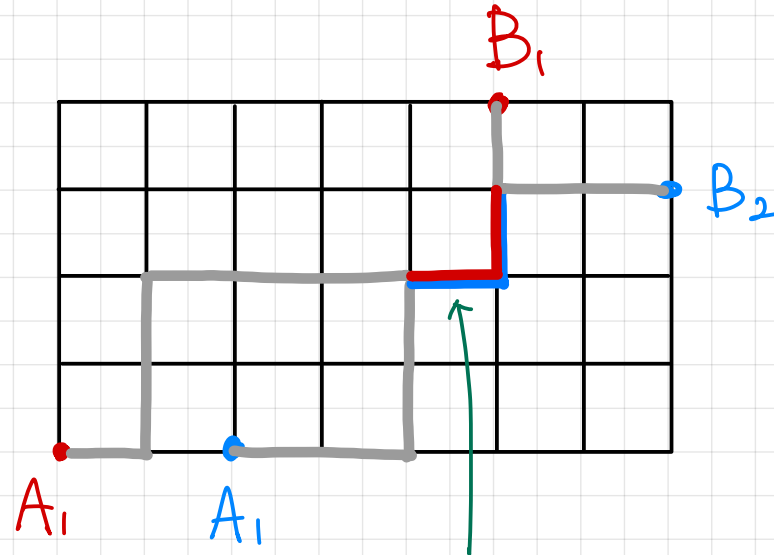
—

?

Lattice Path

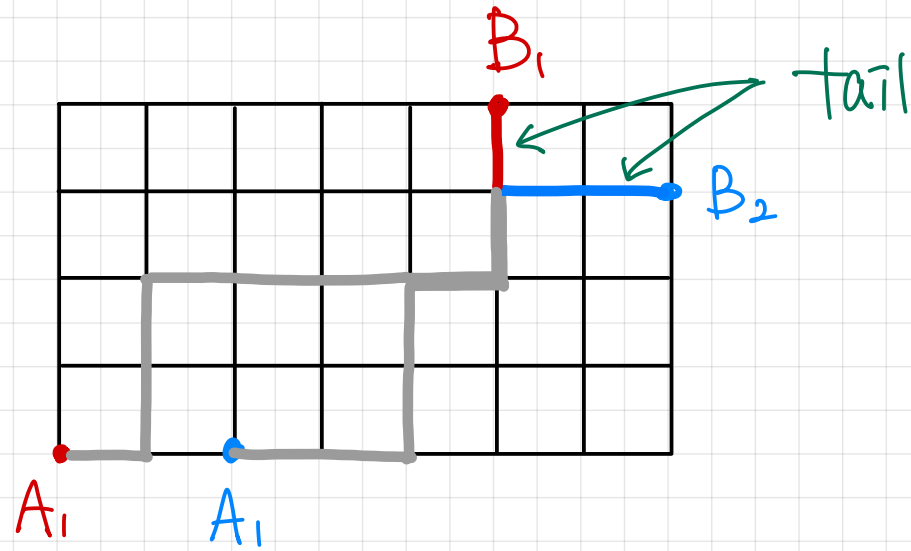


Lattice Path



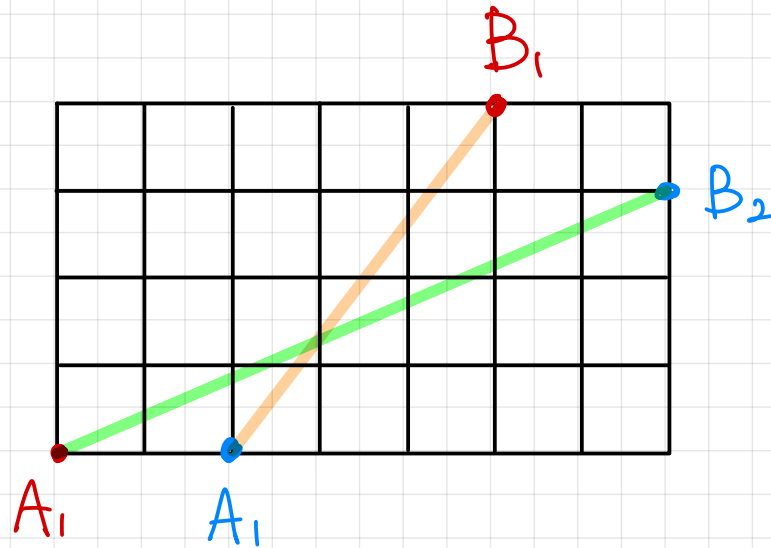
intersecting part
(for the first time)

Lattice Path



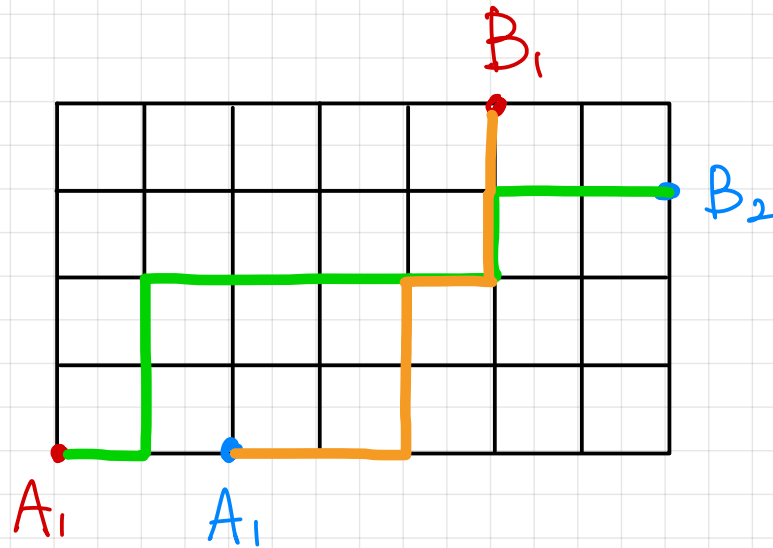
Lattice Path

Think of the paths from A_1 to B_2 and
from A_2 to B_1



Lattice Path

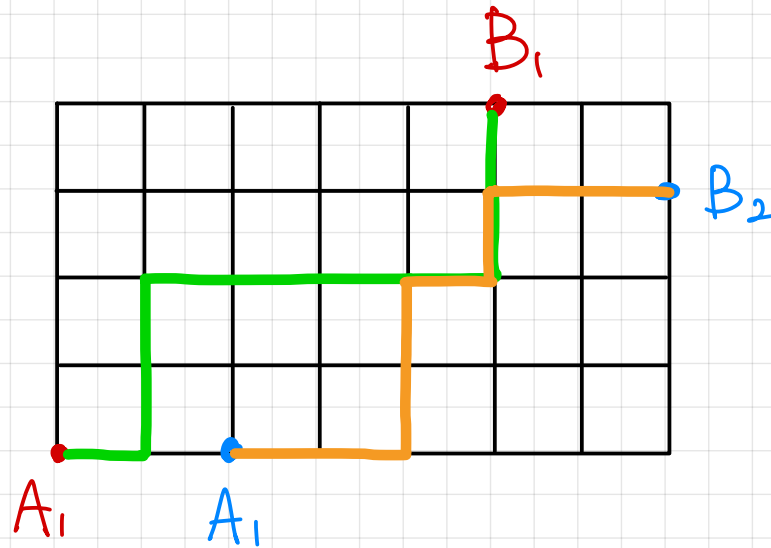
Think of the paths from A_1 to B_2 and
from A_2 to B_1



→ Now, swap the tail!

Lattice Path

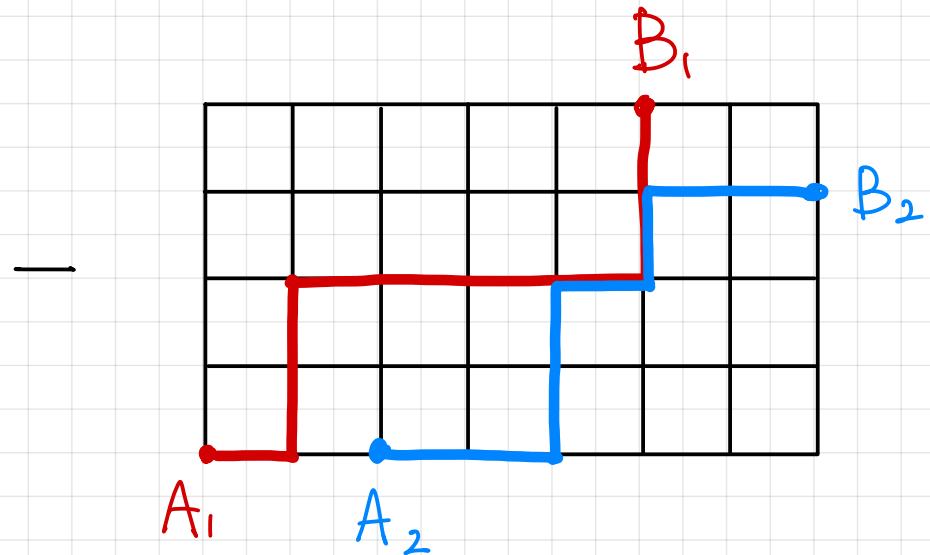
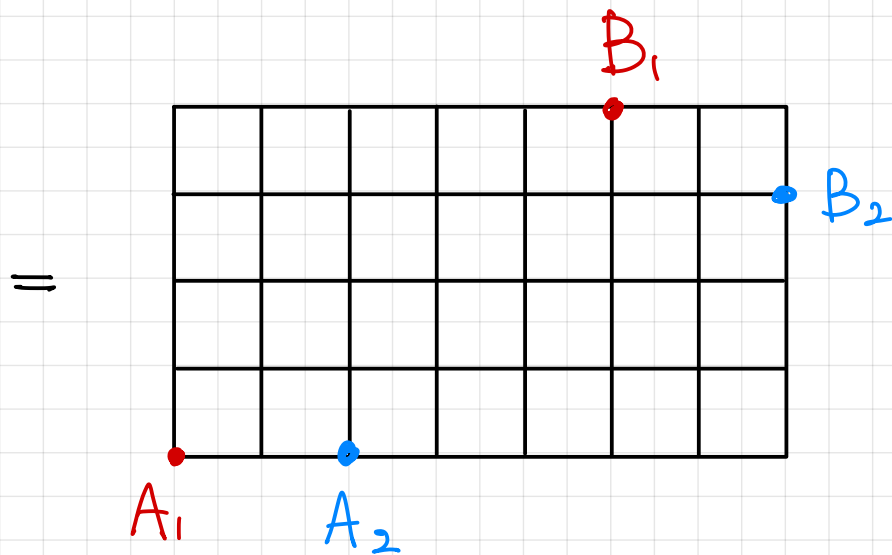
They become the paths from A_1 to B_1 and
from A_2 to B_2
which intersect each other.



Lattice Path

Q. How many such pairs?

A. (# Total pairs) - (# Intersecting pairs)



=

$$\binom{4+5}{4} \times \binom{3+5}{3}$$

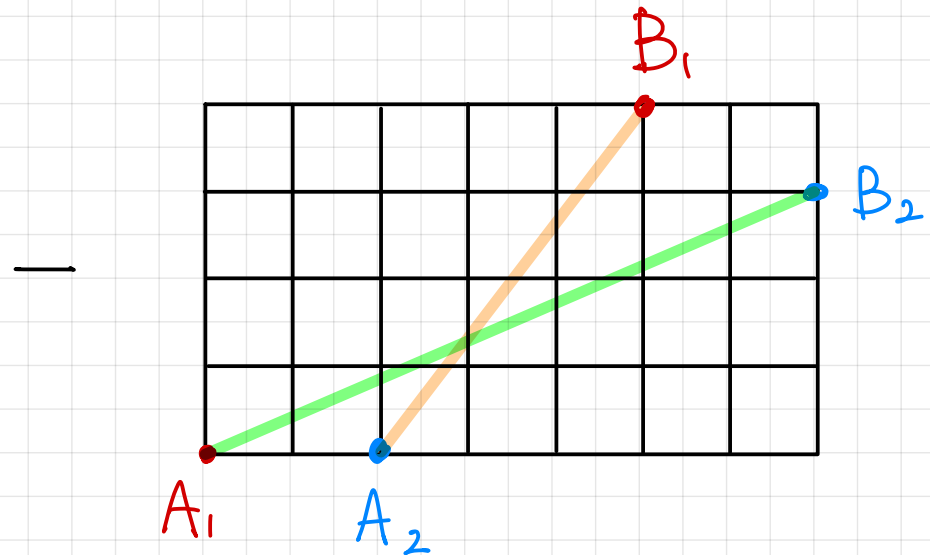
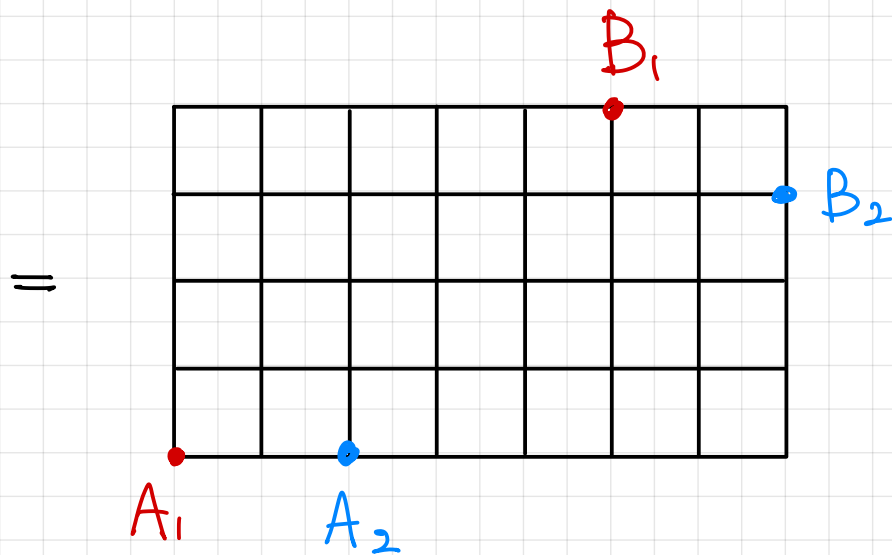
—

?

Lattice Path

Q. How many such pairs?

A. (# Total pairs) - (# Intersecting pairs)



$$= \binom{4+5}{4} \times \binom{3+5}{3} - \binom{3+7}{3} \times \binom{4+3}{4}$$

Lattice Path

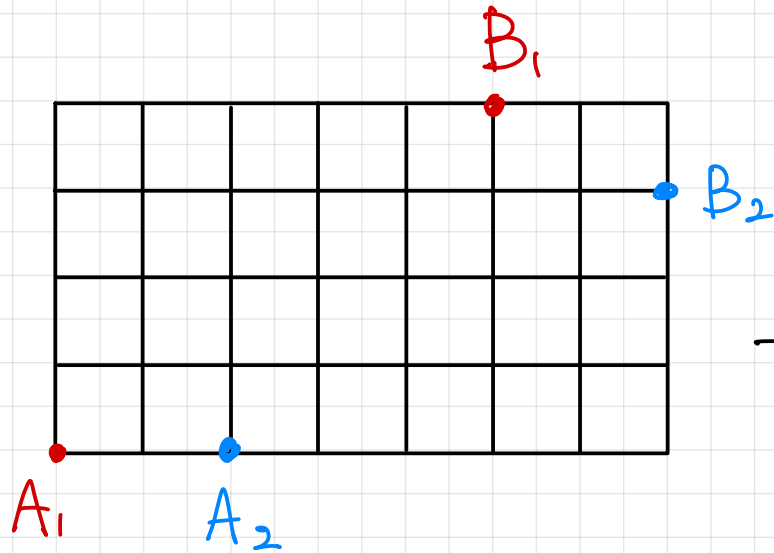
Q. How many such pairs?

A. (#Total pairs) - (#Intersecting pairs)

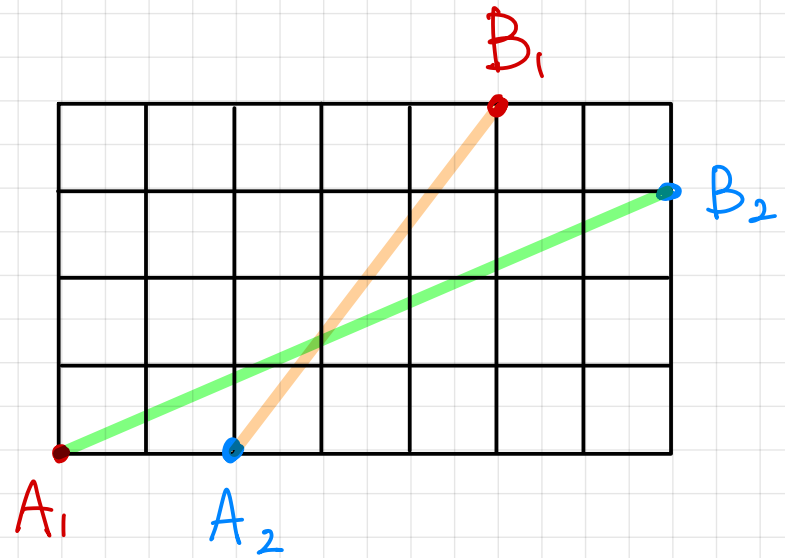
$$= \binom{4+5}{4} \times \binom{3+5}{3} - \binom{3+7}{3} \times \binom{4+3}{4}$$

$$= \det \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}$$

Lattice Path

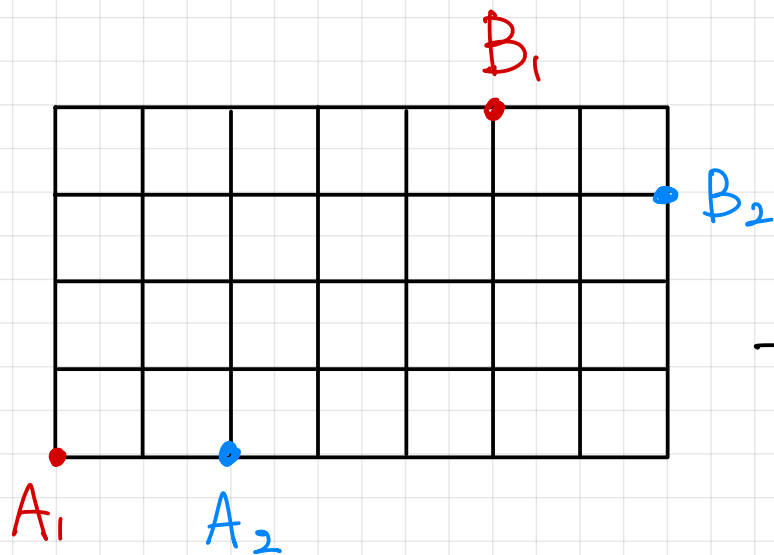


→
tail swap

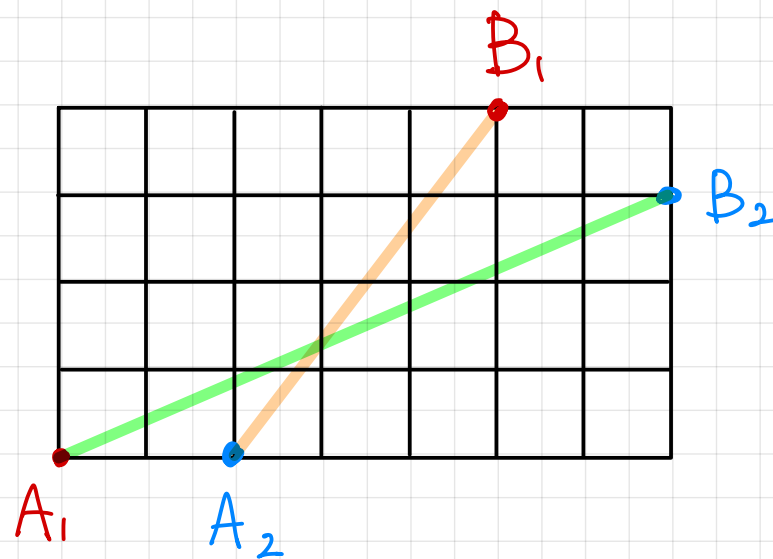


$$\begin{pmatrix} 4+5 \\ 4 \end{pmatrix} \times \begin{pmatrix} 3+5 \\ 3 \end{pmatrix} \xrightarrow[\text{swap}]{\text{destination}} \begin{pmatrix} 3+7 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4+3 \\ 4 \end{pmatrix}$$

Lattice Path



→
tail swap



$$\begin{pmatrix} 4+5 \\ 4 \end{pmatrix}$$

$\times$

$$\begin{pmatrix} 3+5 \\ 3 \end{pmatrix}$$

destination
→
swap.

$$\begin{pmatrix} 3+7 \\ 3 \end{pmatrix}$$

$\times$

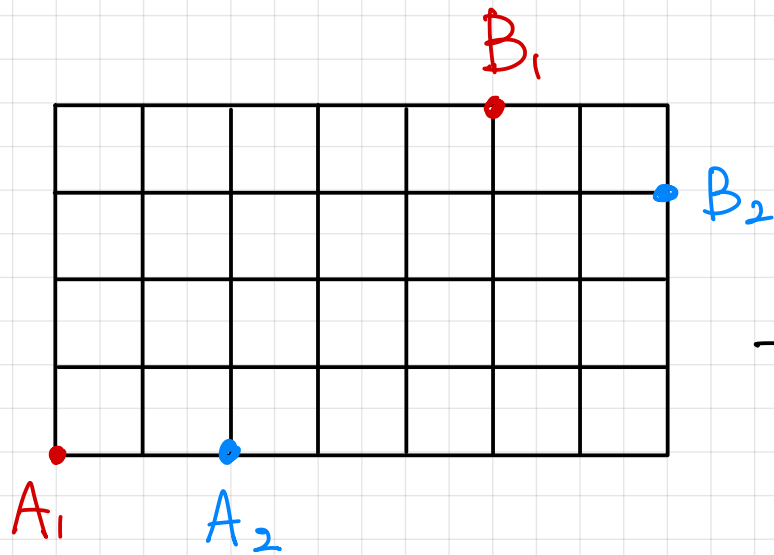
$$\begin{pmatrix} 4+3 \\ 4 \end{pmatrix}$$

↓
define a map σ

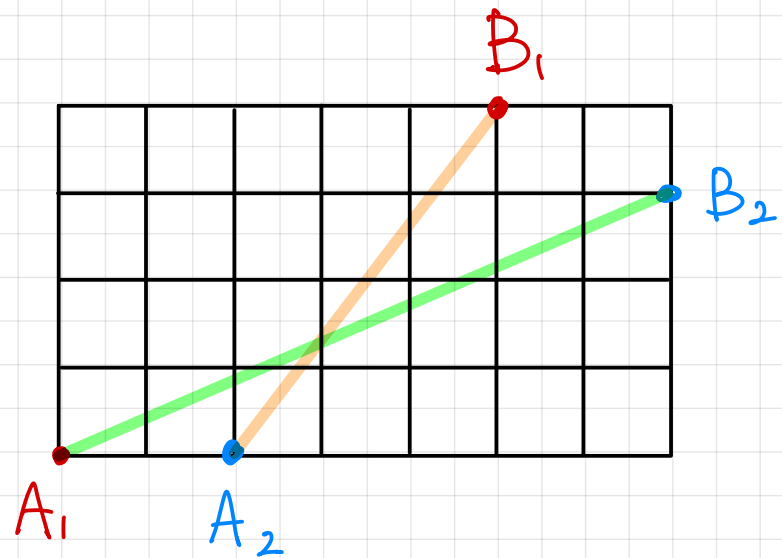
$$\sigma(1) = 2$$

$$\sigma(2) = 1$$

Lattice Path



→
tail swap



$$P_{11} := \text{path } A_1 \rightarrow B_1$$

$$P_{22} := \text{path } A_2 \rightarrow B_2$$

$$\begin{aligned} P_{1\sigma(1)} &:= \text{path } A_1 \rightarrow B_{\sigma(1)} \\ &= \text{path } A_1 \rightarrow B_2 \end{aligned}$$

$$\begin{aligned} P_{2\sigma(2)} &:= \text{path } A_2 \rightarrow B_{\sigma(2)} \\ &= \text{path } A_2 \rightarrow B_1 \end{aligned}$$

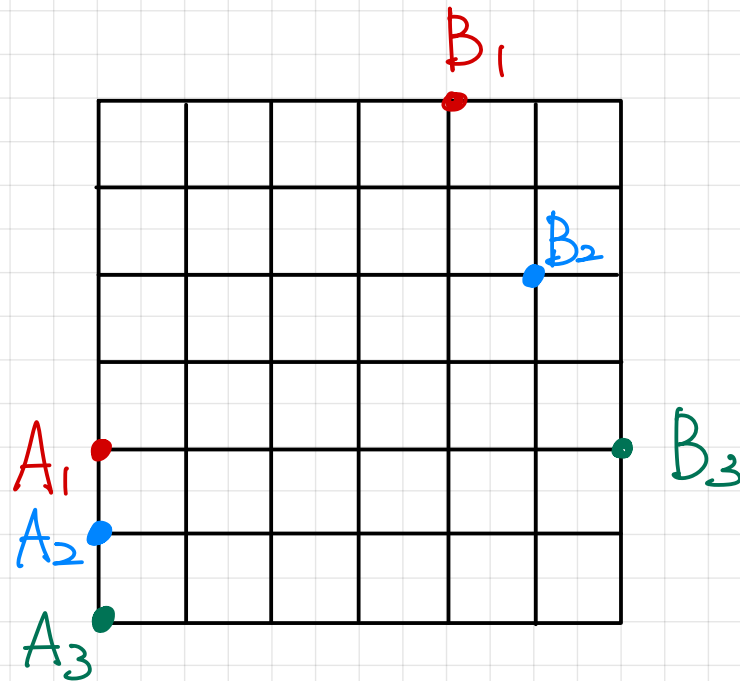
→ σ is called the "permutation"

Lattice Path

$$\det \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} = \sum_{\sigma \in S_2} \underbrace{\text{sgn}(\sigma)}_{\substack{|| \\ (-1)^{(\# \text{swaps})}}} \prod_{i=1}^2 P_{i\sigma(i)}$$

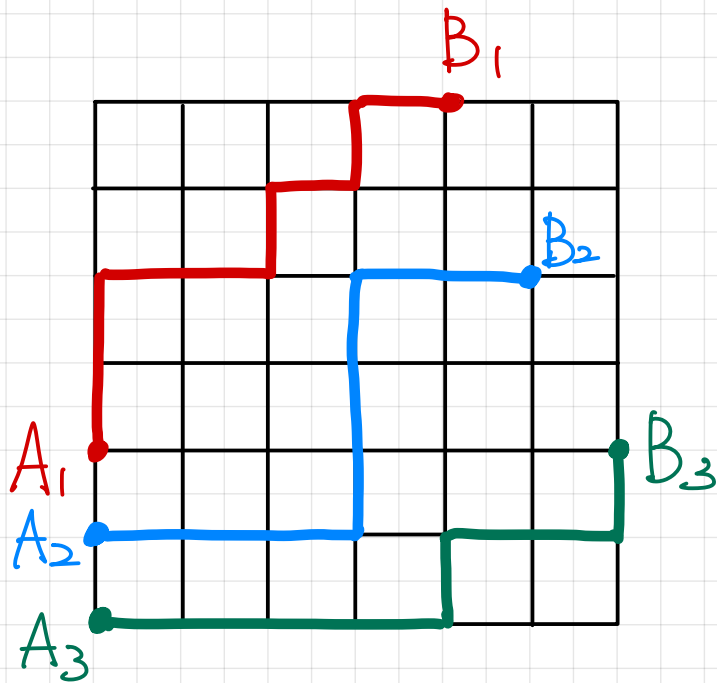
Lattice Path

Q. How about 3 paths?



Lattice Path

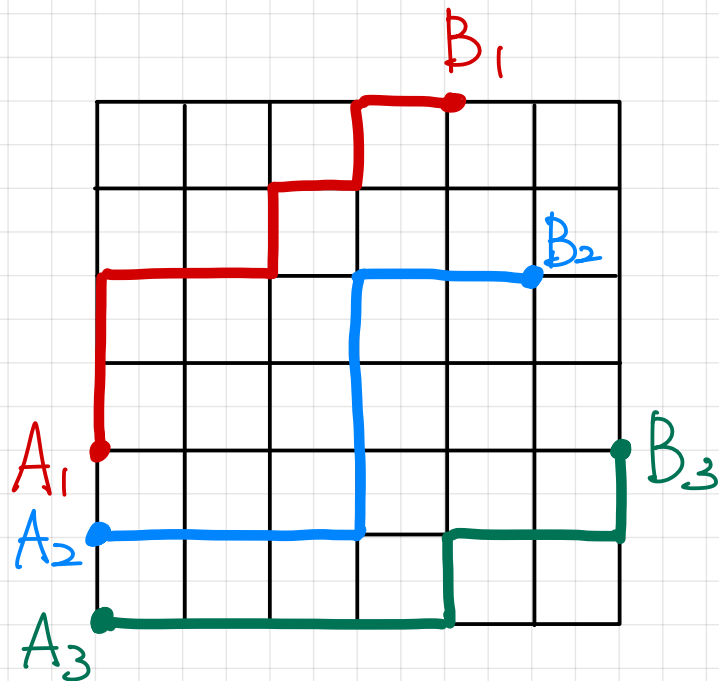
Q. How about 3 paths?



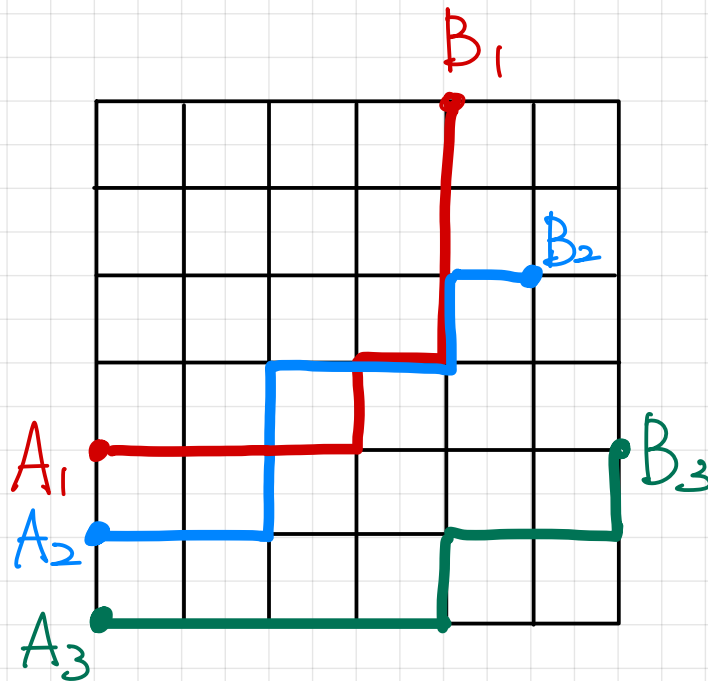
non-intersecting

Lattice Path

Q. How about 3 paths?



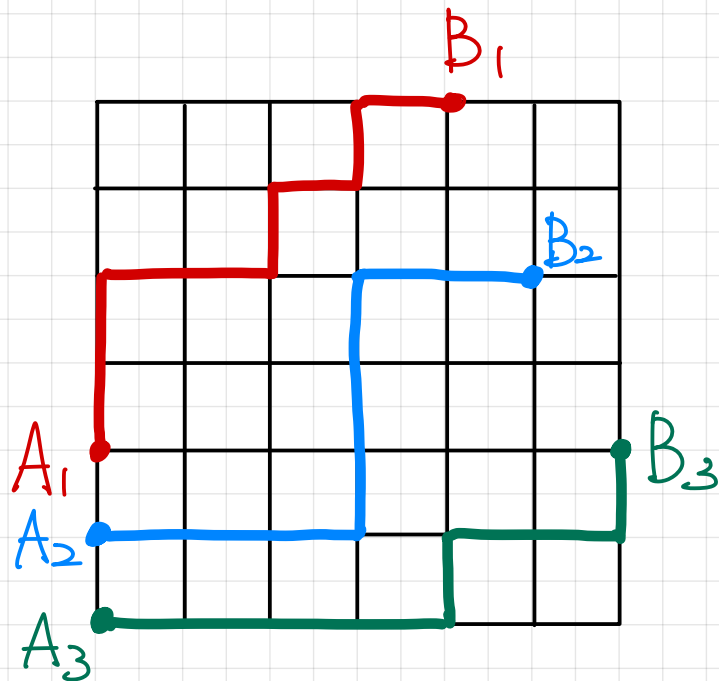
non-intersecting



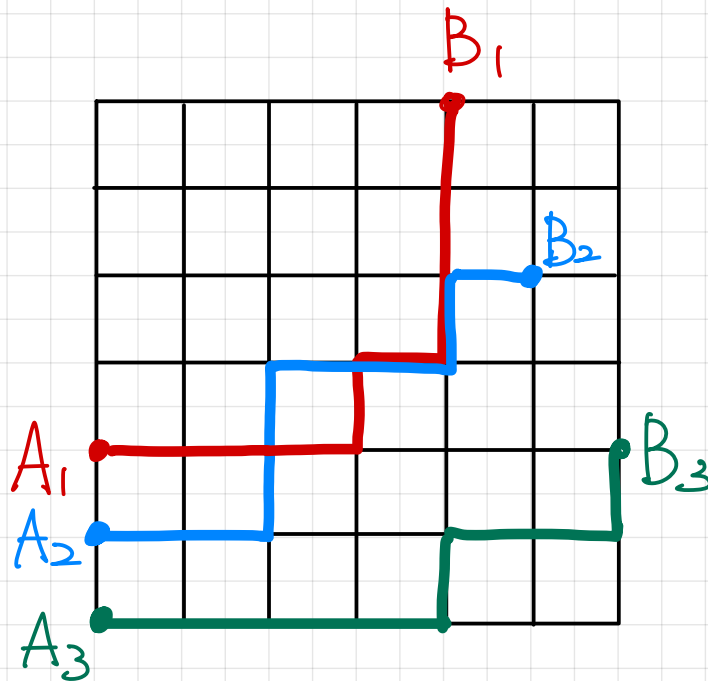
path 1, 2
intersecting

Lattice Path

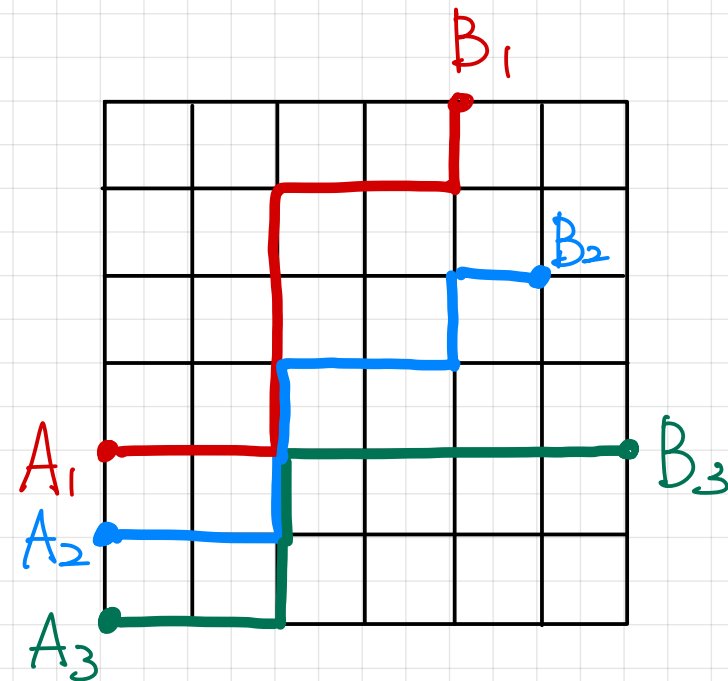
Q. How about 3 paths?



non-intersecting



path 1, 2
intersecting



path 1, 2, 3
intersecting

LGV Lemma

$$\det(M) = \sum_{\sigma \in S_3} \text{sgn}(\sigma) \prod_{i=1}^3 e(A_i, B_{\sigma(i)})$$

σ	$A_1 \rightarrow$	$A_2 \rightarrow$	$A_3 \rightarrow$	# swaps	sign
id	B_1	B_2	B_3	0	+
(23)	B_1	B_3	B_2	1	-
(12)	B_2	B_1	B_3	1	-
(13)	B_3	B_2	B_1	1	-
(123)	B_2	B_3	B_1	2	+
(132)	B_3	B_1	B_2	2	+

For n paths : $\det(M) = (\# \text{non-intersecting})$

Thank You

